



Aqueous acidic solution - ionic liquid biphasic system: An original investigation by diffusion ordered spectroscopy nuclear magnetic resonance

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ARTICLE INFO

Article history:

Received 12 January 2018

Received in revised form 13 August 2018

Accepted 14 August 2018

Available online 18 August 2018

Keywords:

Ionic liquid

NMR 1 and 2 dimensional

Self-diffusion coefficient

ABSTRACT

Transfers of solutes from an aqueous phase to an ionic liquid phase and vice versa has been well studied for years. On the other hand, influence of the transferred solutes on the structure of the ionic liquid phase is not yet evaluated. Therefore, we focus on the aqueous - ionic liquid biphasic system composed of two immiscible phase containing nitric or hydrochloric acid and the $[C_4mim][Tf_2N]$ ionic liquid. They have been yet examined in terms of water and acid solubility in the ionic liquid rich phase. After mixing and separation of the two phases, a part of water, H_3O^+ and nitrates species were transferred into the $[C_4mim][Tf_2N]$ ionic liquid phase, but not chloride anions. The presence of the solutes water, H_3O^+ and nitrates in $[C_4mim][Tf_2N]$ ionic liquid phase could modify local structure of $[C_4mim][Tf_2N]$ ionic liquid. So far, our goal is to evaluate this structural change using several tools. For this, one dimensional 1H , ^{19}F and ^{15}N Nuclear Magnetic Resonance, Nuclear Overhauser effect and Diffusion Ordered Spectroscopy Nuclear Magnetic Resonance were carried out on the $[C_4mim][Tf_2N]$ liquid ionic phase. We observed that some of water molecules were free and formed local microstructure as water pocket inside the ionic liquid frame. The others were hydrogen bonded to $[Tf_2N]^-$ anions via their oxygen atoms from sulfonyl groups. The free water fraction decreased as the concentration of $[HNO_3]_{aq, init}$ increased. In case of $[C_4mim][Tf_2N]$ ionic liquid mixed with hydrochloric acid, the fraction of hydrogen bonded water molecules was more important than with nitric acid.

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1. Introduction

Ionic liquids (ILs) are good alternative to replace classical solvents in liquid-liquid extraction process because of their unique physicochemical properties like high chemical and thermal stability [1], non-flammability and very low vapor pressure [2]. As example, 1-alkyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (henceforth indicated as $[C_nmim][Tf_2N]$, where $n = 2, 4, 6, 8$ and 10), is a typical class of ILs [3–5]. Their capacity as solvents for metal extraction is a proof of the interest and show why they are studied.

Bulk structure of neat $[C_nmim][Tf_2N]$ ($1 \leq n \leq 10$) ionic liquids was intensively studied by small-wide angle X-ray scattering (SWAXS) [6–8], by Infra-Red and Raman spectroscopies [9]. The microscopic scale was characterized by coulombic and Van der Waals interactions between each ion. Hydrogen bond, ions cluster or self-assembled nanostructure built a mesoscopic network [10,11]. For $n \leq 4$ a dispersed phase was described and for $n > 7$ a mesoscopic phase was observed with polar and non-polar domains. Transport properties of 1-alkyl-3-

methylimidazolium $[C_nmim]^+$ ($n = 2, 4, 6, 8$ and 10) cations and bis(trifluoromethanesulfonyl)imide $[Tf_2N]^-$ anions were studied by Diffusion Ordered Spectroscopy Nuclear Magnetic Resonance (DOSY NMR) [12,13]. Cations and anions self-diffusion coefficients D were measured. The D values decreased in order $[C_2mim][Tf_2N] > [C_4mim][Tf_2N] > [C_6mim][Tf_2N] > [C_8mim][Tf_2N] > [C_{10}mim][Tf_2N]$. The imidazolium cations self-diffusion coefficients were more higher than the bis(trifluoromethanesulfonyl)imide anion for $n \leq 6$ and almost equals for $n = 8$ and equal for $n = 10$. The motion of each ion of $[C_nmim][Tf_2N]$ was interconnected with macroscopic physical properties such as viscosity, conductivity and microscopic aspects as intermolecular forces, local microstructure and dynamic heterogeneities of cations. Considering microscopic phenomena, coulombic forces, hydrogen bonding and Van der Waals interactions played their roles and the equilibrium between them changes with the longer of imidazolium ring alkyl chain.

Aqueous - ionic liquid biphasic systems are composed of two immiscible phases as an acidic aqueous phase (nitric, hydrochloric acid ...) and an ionic liquid as $[C_nmim][Tf_2N]$. In our previous work [14–16], we demonstrated the partial transfer of water, H_3O^+ and anion of acid into the ionic liquid phase despite their hydrophobicity. The partial aqueous solubility of the $[C_nmim]^+$ cation and $[Tf_2N]^-$ anion was

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measured. The aqueous – [C_nmim][Tf₂N] ionic liquid biphasic systems are well defined considering equilibrium and transfer of ions between IL and aqueous acidic phases.

With a microscopic point of view, the presence of water and ions might affect local structure of imidazolium based ionic liquid. Sieffert and Wipff [17] and Danten et al. [18] demonstrated by molecular simulation, that water molecules were generally hydrogen-bonded to two [Tf₂N[−]] anions via their oxygen atoms in [C₄mim][Tf₂N] ionic liquid. A part of water was “quasi free” and the other one could form H-bonds with [Tf₂N[−]] anions [19]. Exchange of water molecule from one anion to the other occurred within a few picoseconds. This exchange could not be seen with NMR measurements because the timescale was in the order of milliseconds. Authors calculated that the [C₄mim⁺] cations made weaker H-bond with water than [Tf₂N[−]] anions. Others DFT studies on [C₂mim][Tf₂N] suggested that water molecules was associated with the IL in two forms. One strong single H-bond to the anion while the other hydrogen atom was “quasi free” and with two H-bonds either to two anions or to one anion only (double donor) [19–21]. Kurnia et al. studied the solubility of water into imidazolium based [Tf₂N[−]] ionic liquids by an experimental and theoretical approach. They showed that the solvation of water in ILs occur mainly between IL anion [Tf₂N[−]] and water for $n \leq 6$. For $6 < n < 12$, water interacted with [Tf₂N[−]] anion and IL cation in the polar region [22,23]. To the best of our knowledge, no experimental studies by SWAXS have been carried on imidazolium based [Tf₂N[−]] ionic liquids saturated with water. Confined water and hidden water pocket was highlighted by small-angle X-ray scattering (SAXS) and small-angle neutron scattering (SANS) conducted on 1-butyl-3-methylimidazolium nitrate ionic liquid [24].

Transport properties of [C₄mim][Tf₂N] and water system have been studied by molecular dynamics simulations [17,25] and by DOSY NMR [26]. The self-diffusion coefficients variation of each ions [C₄mim⁺] and [Tf₂N[−]] versus water amount was slightly significant until the water saturated the ionic liquid phase. Water's self-diffusion coefficient increased as the amount of water in ILs phase increased. Authors postulated that water could form micro domains in [C₄mim][Tf₂N] phase and supported the idea of free and bond water [26].

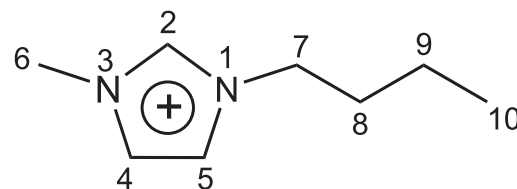
Organic phase of biphasic system, composed of [C₄mim][Tf₂N] contacted with water, nitric and hydrochloric acid solutions of required concentrations, was studied in first time by ¹H, ¹⁹F and ¹⁵N one dimensional NMR. Unfortunately, nuclear Overhauser effect measurements could not be carried on neat [C_nmim][Tf₂N] ionic liquid because of their high viscosity [12] which leads to spin diffusion. By contrast, the transfer of water into ionic liquid phase involved a decrease of ionic liquid phase viscosity [27]. So, in second time new tests of nuclear Overhauser effect measurements were carried on selected samples. In third time, the [C₄mim⁺], [Tf₂N[−]], water and H₃O⁺ ions self-diffusions coefficients were measured in [C₄mim][Tf₂N] phase. This is the first time that the results of the three methods have been collected to understand the behavior of water molecules, H₃O⁺ and nitrate species in the [C₄mim][Tf₂N] ionic liquid phase.

2. Experimental

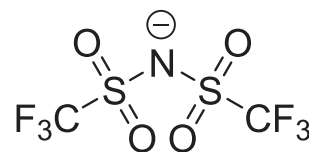
2.1. Materials

High purity grade (99.9%) ionic liquid 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (henceforth indicated as [C₄mim][Tf₂N]) was purchased from Solvionic (Toulouse, France). Chemical structure of the ionic liquid investigated in present work and the atom numbering of the cation are depicted in Fig. 1.

Deuterium oxide (D₂O), deuterated dimethyl sulfoxide (C₂D₆OS, [D₆]DMSO) and deuterio-nitric acid (DNO₃, 65 wt%) of analytical grade (Sigma-Aldrich) were of 99 atom % purity of deuterium. Concentrated nitric acid (HNO₃, 65 wt%) was purchased from Merck and hydrochloric acid (HCl 35 wt%) from Sigma-Aldrich. Sodium trifluoroacetate (CF₃COONa, 98%) was obtained from Alfa Aesar. Deionized water



1-butyl-3-methylimidazolium, [C₄mim⁺]



bis(trifluoromethanesulfonyl)imide, [Tf₂N[−]]

Fig. 1. Chemical structure of ionic liquid cation and anion used in present work.

(ELGA PURELAB DV25) with a specific resistance of 18.2 MΩ·cm was used to prepare the nitric and hydrochloric acid solutions.

2.2. Sample preparation

An aliquot of aqueous nitric or hydrochloric acid solution of required concentration was placed in an Eppendorf test tube (2 mL) together with neat [C₄mim][Tf₂N] (organic-to-aqueous phase ratio of 1:1) [28]. Volume of ionic liquid was accurately evaluated from the mass by using the density of dry [C₄mim][Tf₂N] (1.43 g·cm^{−3} at 25 °C [29]). Aqueous solutions containing 4.0, 5.3 and 7.2 M nitric acid ([HNO₃]_{aq, init}) and 2.0 and 4.0 M deuterio-nitric acid were prepared by diluting HNO₃ or DNO₃ with D₂O, respectively. Aqueous solution containing 7.15 M hydrochloric acid was prepared by diluting concentrated HCl with D₂O ([HCl]_{aq, init}). 1.144 g of [C₄mim][Tf₂N] corresponding 0.8 mL was precisely weighted (Sartorius, BP 2215 model with an accuracy of ±0.1 mg) and 0.8 mL of nitric or hydrochloric acid solution was added. Tubes were agitated during 3 h on a mechanical shaker (IKA vibrax VXR basic) at $t = 23 \pm 2$ °C, at atmospheric pressure. Then, the samples were centrifuged for 4 min at 7000 rpm (Micro Star 12, VWR) and phases were separated. ILs phases were collected to be analyzed by NMR.

NMR measurements required a double sample tube was used in order to separate the IL sample and the lock solvent. The coaxial inner tube (1.4 mm I.D., 1.7 mm O.D.) was filled with a small volume of ionic liquid (0.1 mL) and it was carefully sealed [30,31]. Then, the capillary tube was inserted into outer tube (regular 5 mm NMR glass tube) along with the deuterated solvent, namely with [D₆]DMSO/CF₃COONa. Residual internal peaks of [D₆]DMSO (2.50 ppm) was used as ¹H chemical shifts reference [32, 33], while the ¹⁹F chemical shifts were referenced to the peak of CF₃COONa dissolved in [D₆]DMSO (−74.00 ppm) [34]. Quantity of CF₃COONa in [D₆]DMSO was approximately equal to the mole number of [Tf₂N[−]] anions in the capillary insert.

2.3. NMR measurements

NMR acquisition method was described in our previous work [12] on neat ionic liquid. The same method was used on biphasic samples. Briefly, the one dimensional spectra of ¹H and ¹⁹F were recorded on a Bruker 300 MHz instrument with a 5 mm PABBO BB-1H/D Z-GRD probe. The chemical shift uncertainty measurement was ±0.01 ppm.

DOSY NMR experiments were carried out on a Bruker 600 MHz spectrometer equipped with a high strength z gradient probe DOTY Scientific, developing a pulse field gradient of 500 G/cm. Self-diffusion coefficients, *D*, of each species were measured with pulsed field gradient

NMR experiments. Pulse sequence used was a stimulated echo with bipolar gradients and a longitudinal eddy current delay (BPLED) [35–37].

The $[\text{C}_4\text{mim}^+]$ self-diffusion coefficient was measured by ^1H NMR and named D_+ and the one of water D_{HOD} . The $[\text{Tf}_2\text{N}^-]$ self-diffusion coefficient was measured via ^{19}F DOSY NMR and named D_- . Thus, the self-diffusion coefficients of IL constituent anion and cation were determined individually. The experimental errors in D were estimated to be 10%. R was defined by the ratio D_+ divided by D_- [12].

One dimensional ^{15}N spectra, two dimensional ^1H – ^{15}N Heteronuclear Multiple Quantum Coherence HMQC and ^1H – ^1H Nuclear Overhauser Enhancement Spectroscopy NOESY spectra were recorded on a Bruker 500 MHz avance III HD instrument equipped with a 5 mm cryo probe BBO on selected samples. All NMR measurements were carried out at 298 K.

2.4. Other analytical measurements

The protons' aqueous concentration was determined by acid-base titration (SCHOTT instrument Titroline easy) using NaOH standard solutions (Carlo Erba). Total water content in the ionic liquid phase was determined by the coulometric Karl-Fisher technique (Hydranal®-Coulomat E, Fluka) [12]. Measurements uncertainties were 2% in the molar concentration of H^+ or D^+ in aqueous phase and 5% on the molar fraction of water in ILs phases.

3. Results and discussion

NMR measurements for $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid phase contacted with aqueous acidic phases are divided in three parts. First part shows the 1-dimensional NMR results of ^1H , ^{19}F and ^{15}N spin. Second part describes the NOE effects observed on a selection of samples. Third part presents the results of DOSY NMR.

3.1. Sample composition

Four types of liquid-liquid extraction experiments were performed by mixing neat $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ organic phase with aqueous solution containing:

- 0.50 $\text{H}_2\text{O}/0.50 \text{D}_2\text{O}$ in v/v named HOD
- Deuterated nitric acid in D_2O at 1.99 and 3.91 M
- Nitric acid in D_2O at 4.00, 5.28 and 7.22 M
- Hydrochloric acid in D_2O at 7.14 M.

In the aqueous phase, only primary isotopic exchange processes are taken in account [38] (before contact with ionic liquid phase), the equilibria can be described by the following reactions:



The dissociation of nitric and hydrochloric acid is taken in account:



Speciation in aqueous solution of nitric acid was studied by Raman spectroscopy [39]. Three species of nitric acid were highlighted, free nitrate ions, ion pairs and neutral HNO_3 molecules. For $[\text{HNO}_3]_{\text{aq, init}} < 2 \text{ M}$ free nitrates species were predominant, in a range of 3–8 M the quantity of ion pairs is double as many than HNO_3 molecules. Table SI 2 gives the molar concentration of each nitric acid species in aqueous solution for $[\text{HNO}_3]_{\text{aq, init}} = 4, 5$ and 7.12 M .

Thus H_2O , D_2O , HOD , H_2DO^+ , HNO_3 , NO_3^- , Cl^- and H_3O^+ existed in aqueous phase and equilibria were reached very quickly [40]. The use

of H_2O only, yields to a broad signal of water which could overlap others signals and disturbed the NMR measurements. The biphasic system is noted as $\text{HOD}/\text{HNO}_3//[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$, or $\text{HOD}/\text{HCl}/[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ following the aqueous phase initial composition. The two immiscible phases, acidic aqueous and ionic liquid were contacted and separated.

Despite the $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid hydrophobicity, water saturates the organic phase [41]. In our previous works [14,15], the partial solubility of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ in aqueous nitric acid phase and the transfer of water with a small amount of HNO_3 in IL phase was demonstrated. Samples were chosen according to the solubility curve of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ in aqueous nitric acid solution. Neat $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid contacted with hydrochloric acid solution was saturated with water and only a tiny part of H_3O^+ was transferred into the ionic liquid phase. Chloride anions were not extracted by the organic phase [16]. From these known results, the composition of aqueous phases and mole fraction of all entities presented in ILs phase after liquid-liquid extraction are described in Table 1.

The use of nitric acid solutions with increasing concentrations promoted the transfer of water molecules, H_3O^+ and nitrates species in the ionic liquid phase during mixing between the two phases. In our previous work [15] quantities transferred into the ionic liquid phase was determined as a function of the acid nitric initial concentration in aqueous phase. Table 1 displays the values of initial and final concentration of HNO_3 or DNO_3 in deuterated aqueous phase, the HOD , H^+/D^+ and nitrate species ions concentration transferred in IL phase. The H_3O^+ ion amount into ionic liquid phase is slightly lower than the nitrates species and one nitrate anion is extracted with three molecules of HOD by $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid. The water amount in the ionic liquid phase increases drastically when $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ is mixed with increasing nitric acid concentrations. For $[\text{HNO}_3]_{\text{aq, init}} = 7.22 \text{ M}$, the HOD mole fraction extracted reaches almost $[\text{C}_4\text{mim}^+]$ and $[\text{Tf}_2\text{N}^-]$ anion mole fraction. Considering the molar fraction, the main species in ionic liquid phases are $[\text{C}_4\text{mim}^+]$ cation, $[\text{Tf}_2\text{N}^-]$ anion and HOD . The HOD concentration is *ca* 5 times higher than the mole fraction of H_3O^+ and acid anion. Minor species are H_3O^+ and nitrates whatever initial nitric acid concentration. In the case of hydrochloric acid, the ionic liquid phase contains only water, $[\text{C}_4\text{mim}^+]$ cations and $[\text{Tf}_2\text{N}^-]$ anions.

3.2. ^1H , ^{19}F and ^{15}N 1D NMR measurements

^1H 1D NMR spectra of ionic liquid phases contacted with several nitric acid solutions, were recorded. The $[\text{D}_6]\text{DMSO}$ residual Internal peak (2.50 ppm) was used as ^1H chemicals shifts reference. Fig. 2 shows the ^1H 1D spectra of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid phase mixed with HNO_3 in deuterated water for each initial concentration.

$[\text{C}_4\text{mim}^+]$ cation chemicals shifts are given in Table SI 1. For all samples, $[\text{C}_4\text{mim}^+]$ proton chemical shift stayed constant. Water ^1H NMR signal which was transferred into ionic liquid phase is located at $\approx 4.00 \text{ ppm}$ [42,43]. It is slightly shielded in comparison to the usual water chemical shift at 4.79 ppm [33]. An additional peak appears at 2.38 ppm for the sample $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid saturated with HOD . This peak doesn't exist with neat $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ [12]. The $[\text{C}_2\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid ^1H spectrum containing 1.0 M of water showed a peak at 2.82 ppm [44]. In the case of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid mixed with HNO_3 in deuterated phases with initial concentration at 4.00, 5.28 and 7.22 M, the additional proton peak appears at 5.12, 5.88, and 6.67 ppm respectively as Fig. 2 shows. In the case of ionic liquid phase contacted with HCl 7.14 M the chemical shift of this proton is 5.40 ppm as shown in Fig. SI 1.

A part of water which was extracted by ionic liquid phase, creates local microstructure or water pocket and could be considered as 'quasi free' [26,45]. Water could also be considered as an inclusion in the ionic liquid $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ supramolecular network [46]. Its ^1H NMR chemical shift is 4 ppm. The other part of water is bonded and its chemical shift varies with the amount of water extracted by $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid. Thus, the protons signals at 5.12, 5.88, 6.67 and 5.40 ppm

Table 1

Samples composition of aqueous and ILs phases.

Composition of aqueous phases			Composition of [C ₄ mim][Tf ₂ N] phases [14,15]				
Acid used	[Acid] _{aq, init} M	[H ⁺ /D ⁺] _{aq, eq} M	Mole fraction of H ⁺ /D ⁺	Mole fraction of acid anion	Mole fraction of HOD	Mole fraction of [C ₄ mim ⁺]	Mole fraction of [Tf ₂ N [−]]
/	0	pH = 6.01	/	/	0.13	0.43	0.43
DNO ₃ in D ₂ O	1.99	1.98	/	/	0.13	0.43	0.43
	3.91	3.62	0.03	0.05	0.16	0.37	0.37
HNO ₃ in D ₂ O	4.00	3.75	0.03	0.05	0.16	0.37	0.37
	5.28	4.91	0.04	0.06	0.19	0.35	0.35
	7.22	6.71	0.05	0.08	0.25	0.31	0.31
HCl in D ₂ O	7.14	6.98	0.02	/	0.13	0.43	0.43

for [HNO₃]_{aq, init} = 4.00, 5.28, 7.22 M and [HCl]_{aq, init} = 7.14 M, are assigned to bound water. Under this hypothesis, exchanges described by the equilibria (1), (2), (3) or (4) could be considered inside water pocket with several equilibria constants. One more reaction is added by the exchange between water free and bonded inside ionic liquid phase:



The nitrates and H₃O⁺ amount are minor compared to the one of water. So equilibria (1) and (5) are predominant.

¹⁹F 1D NMR spectra on the same ionic liquid phases were recorded. The reference was CF₃COONa dissolved in DMSO. The —CF₃ chemical shift was set at −74.00 ppm [34]. The —CF₃ chemical shift from [Tf₂N[−]] anion was ca. −80.60 ppm and the fluorine signal was a singlet for each sample. There were no displacements of the —CF₃ chemical shift from [Tf₂N[−]] anion. They were not disturbed by the addition of water, H₃O⁺ or nitrates ions into ionic liquid phase due to their incapacity to make H-bonds.

¹⁵N 1D NMR measurements were carried out on ionic liquid phase from HOD/[C₄mim][Tf₂N] and HOD/HNO₃/[C₄mim][Tf₂N] with [HNO₃]_{aq, init} = 7.22 M biphasic system as well as Heteronuclear Multiple Quantum Correlation (HMQC) experiments to assign nitrogen of imidazolium ring. Spectra are shown in Figs. SI 2, SI 3 and SI 4. Nitrogen signals are well identified and an additional signal appears for the ionic liquid phase mixed with HNO₃ 7.22 M due to the partial transfer of nitrate into it. Table 2 depicts values of ¹⁵N chemical shift. The low value of the magnetogyric ratio [47] and the very slow relaxation time of ¹⁵N [34] imply very weak signals. It is not possible to measure peak area and to perform NOE measurements.

Table 2¹⁵N chemical shift of the [C₄mim][Tf₂N] ionic liquid phase contacted with [HNO₃]_{aq, init} = 0 and 7.22 M.

[C ₄ mim][Tf ₂ N] ionic liquid phase	δ ppm ¹⁵ N nitrates species	δ ppm ¹⁵ N(1)	δ ppm ¹⁵ N(3)	δ ppm ¹⁵ N (Tf ₂ N [−])
[HNO ₃] _{aq, init} = 0 M	/	184	171	140
[HNO ₃] _{aq, init} = 7.22 M	347	184	171	139

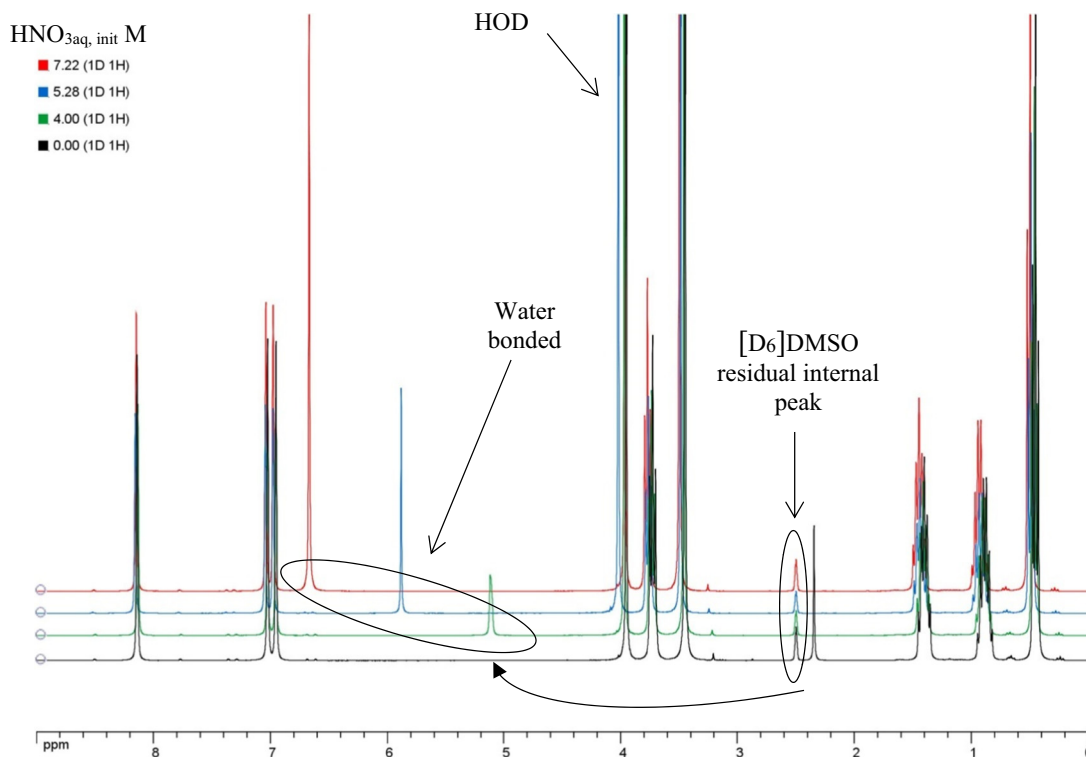


Fig. 2. ¹H 1D spectra of ionic liquid phase from HOD/HNO₃/[C₄mim][Tf₂N] biphasic system. [HNO₃]_{aq, init} = 0.00 M (black), 4.00 M (green), 5.28 M (blue) and 7.22 M (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

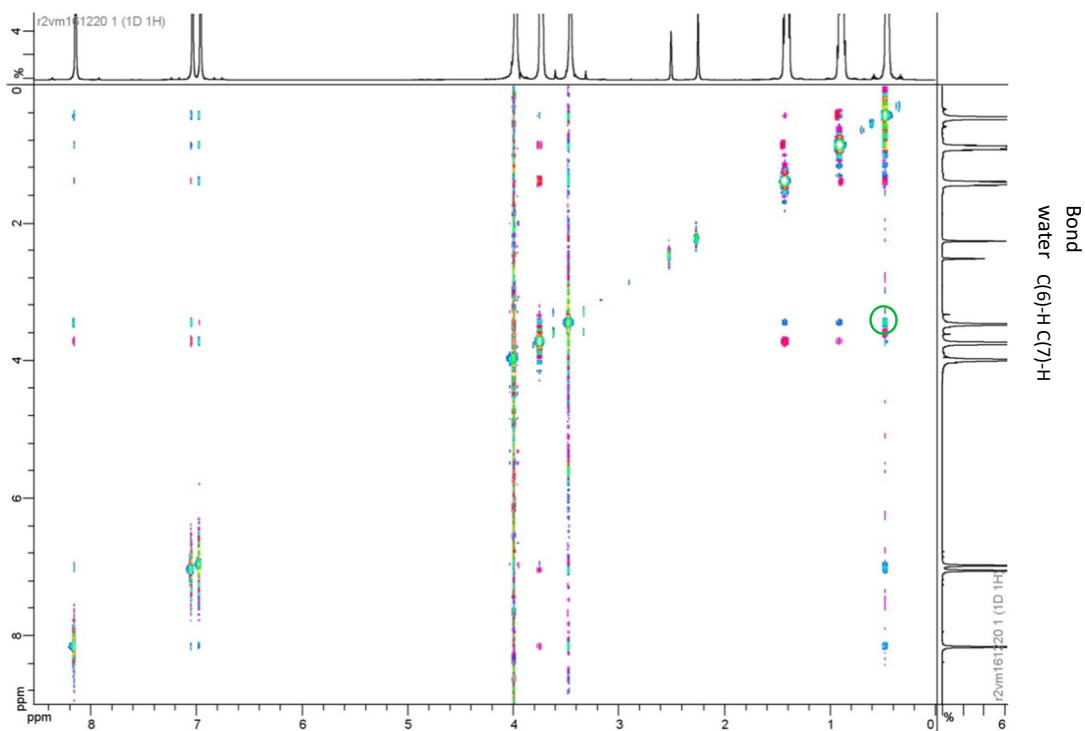


Fig. 3. ^1H – ^1H NOESY spectrum of ionic liquid phase from $\text{HOD}/[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ biphasic system. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

3.3. Nuclear Overhauser effect

In our previous work [12], ^1H – ^1H NOESY and ^1H – ^{19}F HOESY NMR spectra were recorded on neat ionic liquid phase. The spin diffusion

through the molecule due to the high viscosity of the ionic liquid prevented the realization of measurements in NOE NMR [48]. By transfer of water into ionic liquid phase, the viscosity decreased [27]. So new tests were carried on ionic liquid phase from $\text{HOD}/[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ and

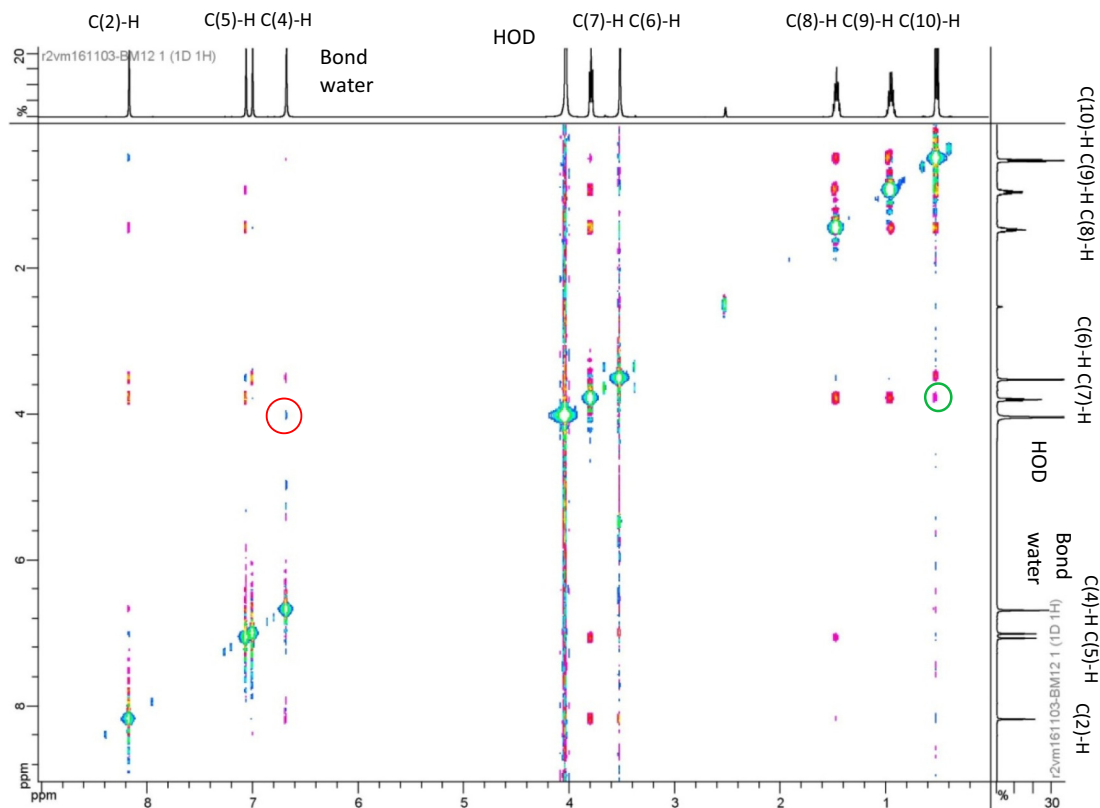


Fig. 4. ^1H – ^1H NOESY spectrum of ionic liquid phase from $\text{HOD}/\text{HNO}_3/[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ biphasic system. The HNO_3 initial concentration is 7.22 M. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

HOD/HNO₃/[C₄mim][Tf₂N] with [HNO₃]_{aq, init} = 7.22 M biphasic system. Figs. 3 and 4 shows ¹H–¹H NOESY spectra with correlation cross peak of cation spin and HOD free or not. The top horizontal and vertical inserts show ‘classic’ ¹H spectra of [C₄mim⁺] cation and atom numbering. Positive levels are in blue and green, negative levels are in red. Plots located on the diagonal straight line are positives and on either side of it, plots are positives and negatives. In the case of a system without spin diffusion, all signals outside the diagonal are signs contrary to those situated on the diagonal. So, diffusion spin seems to be always significant in [C₄mim][Tf₂N] ionic liquid phase saturated with HOD or mixed with [HNO₃]_{aq, init} = 7.22 M and it is impossible to calculate inter molecular distance.

Nevertheless, in the case of [C₄mim][Tf₂N] ionic liquid saturated with HOD, ¹H C(4)–H, ¹H C(5)–H and ¹H C(2)–H correlates with ¹H of alkyl chain, ¹H C(6)–H and ¹H C(7)–H. Another interesting correlation shows the spatial proximity between ¹H C(6)–H and ¹H C(10)–H (green circle in Fig. 3). NOE measurements exhibit a head – to – tail rather than head – to – head aggregation motives of [C₄mim⁺] cations [49]. There is no correlation observed between the bound water proton and the [C₄mim⁺] cation proton. Fig. 5 describes a schematic view of [C₄mim][Tf₂N] ionic liquid frame saturated with water.

In the case of HOD/HNO₃/[C₄mim][Tf₂N] with [HNO₃]_{aq, init} = 7.22 M biphasic system, as shown in Fig. 4, NOE cross-peaks (green circle) are clearly detected between ¹H C(6)–H and ¹H C(10)–H. In other way this signal indicates a spatial proximity between N–CH₃ and methyl from the alkyl chain of [C₄mim⁺] cation. Intensity of cross-peak between imidazolium rings ¹H C(2)–H, ¹H C(5)–H, ¹H C(4)–H and alkyl chain ¹H C(8)–H, ¹H C(9)–H, ¹H C(10)–H is less important than with ionic liquid phase saturated with water. That means the distance between the [C₄mim⁺] cations imidazolium rings increases slightly. The [C₄mim⁺] cations seem to be placed in single file and water inserts between imidazolium rings as shown the Fig. 6.

The linked water proton at 6.67 ppm correlates slightly with ¹H C(10)–H and ¹H C(6)–H. However, a correlation peak is clearly detected between proton from bound water and from HOD (red circle in Fig. 4). This shows exchange between linked water and free water as described by reaction (5).

In both cases, ¹H C(7)–H correlates with ¹H C(8)–H, ¹H C(9)–H, ¹H C(10)–H of the imidazolium alkyl chain. The cross peak intensity decreases along the alkyl chain, as Figs. 3 and 4 shows. So, the alkyl

chain stays stretch instead of curling up whatever the water mole fraction in ionic liquid phase.

The loss of correlation between the imidazolium ring protons means an increase in the intermolecular distance between them. Nevertheless, there remains a spatial proximity between the methyl group N–CH₃ proton and those of methyl group connected at the end of carbon chain. The water mole fraction increasing and the partial transfer of ions H₃O⁺ and NO₃[−] cause [C₄mim][Tf₂N] ionic liquid network to swell while retains the head – to – tail pattern of [C₄mim⁺] cation [23].

In the case of [C₄mim][Tf₂N] ionic liquid saturated with water, no correlation has been observed between bound water and cation. On the other hand, in the case of ionic liquid mixed with [HNO₃]_{aq, init} = 7.22 M, correlation peaks are observed between bound water and [C₄mim⁺] methyl groups protons. This implies a gradual solvation of the cations with water molecules. Moreover, a correlation cross-peak is observed between bound water and HOD (red circle in Fig. 4). This implies a spatial proximity of <5 Å between HOD and linked water. This correlation reinforces the idea of exchange between bound water and free water as described in Eq. (5).

3.4. Transport properties of ionic species and ionic interactions

Diffusion NMR measurements were carried out on ionic liquid phase contacted with HOD, nitric or hydrochloric acid. The [C₄mim⁺] cations self-diffusions D_+ , [Tf₂N[−]] anions D_- and values of ratio $R = D_+/D_-$ are reported in Table 3. All self-diffusion coefficients D_+ and D_- in ILs phases from biphasic samples are higher than pure and dried [C₄mim][Tf₂N] ($D_+ = 3.00 \cdot 10^{-11}$ and $D_- = 2.10 \cdot 10^{-11}$ m² s^{−1}) [12]. As can be seen in Table 3, the D_+ values are always higher than those of D_- regardless of the initial acid concentration in the aqueous phase. The R value with pure and dried [C₄mim][Tf₂N] is 1.43 [12]. The ratio R decreases with the increase of the nitric acid initial concentration. In the case of [C₄mim][Tf₂N] ionic liquid mixed with [HCl]_{aq, init} = 7.14 M, the ratio R is 1.28, minor value than for neat [C₄mim][Tf₂N]. The diffusivities of ionic liquid cation and anion increase upon addition of water, which is rationalized by an overall decrease of viscosity [27,50,51].

The HOD self-diffusion D_{HOD} at $\delta = 4$ ppm is easily detected. Proton signals at 5.12, 5.88 and 6.67 ppm for each nitric acid initial concentration in the deuterated phase and at 5.4 ppm in the case of hydrochloric acid are correctly identified. These chemical shifts are attributed to the

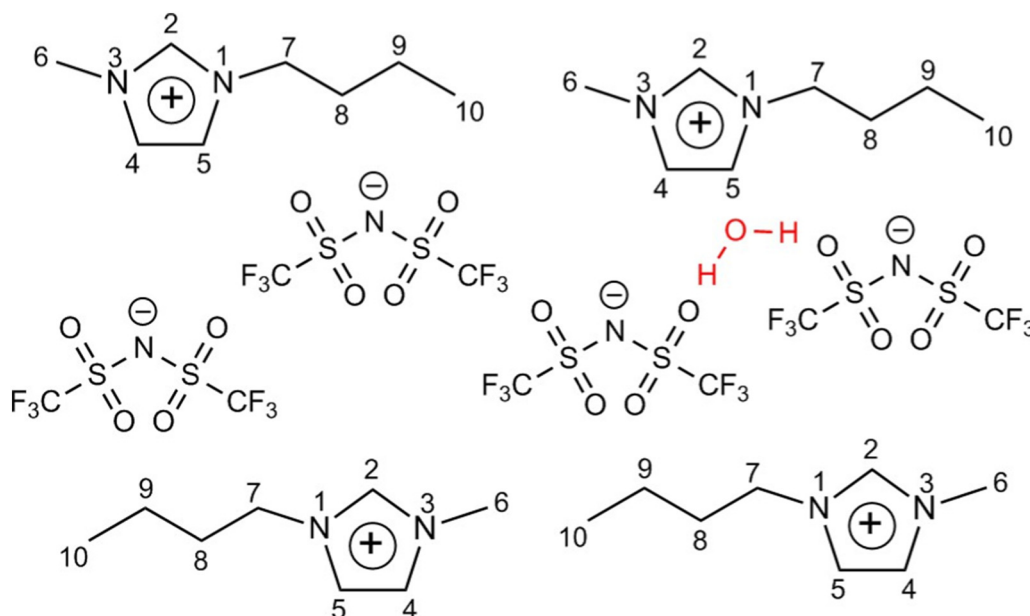


Fig. 5. A schematic view of the [C₄mim][Tf₂N] ionic liquid frame saturated with water.

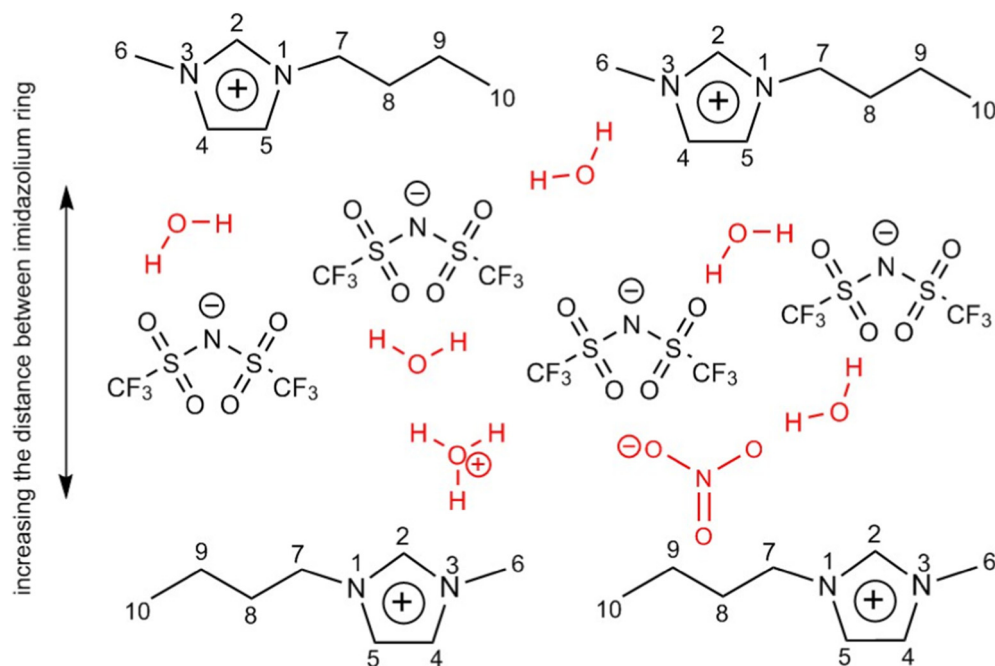


Fig. 6. A schematic view of the $[C_4mim][Tf_2N]$ ionic liquid frame contacted with $[HNO_3]_{aq, init} = 7.22$ M.

protons of bound water molecules. Self-diffusions coefficient $D_{bond\ water}$ is reported in Table 3. Some signals are observed on ^{19}F spectra and self-diffusion coefficient $D_{other\ ^{19}F\ species}$ could be measured. The Table 3 depicts the entire self-diffusion coefficient of all spots identified on 1H and ^{19}F DOSY spectra.

Fig. 7 shows the spectrum of sample $HOD/[C_4mim][Tf_2N]$, in which 1H NMR spectrum of IL cation and ^{19}F NMR spectrum of IL anion are correlated with the variation of self-diffusion D ($\mu m^2\ s^{-1}$). The bottom horizontal axis represents chemical shift of 1H (Fig. 7a) or ^{19}F (Fig. 7b). The top horizontal and vertical inserts show the 'classic' 1H and ^{19}F spectra, and the diffusion peaks, respectively. 1H DOSY spectra and ^{19}F DOSY spectra of $[C_4mim][Tf_2N]$ ionic liquid phase contacted with several concentration of HNO_3 and HCl are displayed in supplementary information (Figs. SI 5 to SI 8).

As seen in Fig. 7, the main signal is due to the $[C_4mim][Tf_2N]$ ionic liquid cations and anions. On the 1H spectrum, HOD signal at 4.00 ppm is on the top. The additional signal at 2.38 ppm is located between the HOD signal and $[C_4mim]^+$ and goes at almost the same speed as species detected by ^{19}F DOSY NMR. Considering ^{19}F spectrum, several signals appear in addition to the main signal of $[Tf_2N]^-$. Two other species composed of $[Tf_2N]^-$ anions move faster than all others species detected in ionic liquid phase, but the signal is tiny and no identification can be realized.

Self-diffusion coefficient of the signal at 2.38 ppm is close to one of a self-diffusion coefficient of other Tf_2N^- species as seen in Table 1. A part of water is engaged in hydrogen bond with $[Tf_2N]^-$ anions via its sulfonyl oxygen group [44]. The additional peak at 2.38 ppm is attributed to $H-O-D$ linked with $[Tf_2N]^-$ anions. $Tf_2N^- \cdots H-O-D \cdots Tf_2N^-$ moved faster than each ion of $[C_4mim][Tf_2N]$ ionic liquid phase. Thus, $[Tf_2N]^-$ anion, although its hydrophobic nature, may form quite strong hydrogen bonds with water by his SO_2 group.

HOD diffuses at the same speed ca. $4.90 \cdot 10^{-10}\ m^2\ s^{-1}$ through the frame of ionic liquid phase regardless of the initial nitric or hydrochloric acid concentration. The self-diffusion coefficient of pure HOD is $2.11 \cdot 10^{-9}\ m^2\ s^{-1}$ [52]. So, HOD is slows down by electrostatic interactions and chemicals exchanges between it and the ionic species presented inside the ionic liquid phase. Furthermore, free water diffuses ca. ten times faster than $[C_4mim]^+$ cation and $[Tf_2N]^-$ anion inside ionic liquid phase. All this elements emphasizes the idea that free water molecules moves in own microstructure inside $[C_4mim][Tf_2N]$ ionic liquid phase [17,26].

Another part of HOD, H_3O^+ or nitrates could interact between each other and both with the $[C_4mim][Tf_2N]$ ionic liquid cation and anion. Cammarata and co-workers [53,54] demonstrated by using infra-red spectroscopy that water molecules interacts stronger with nitrates ions than with $[Tf_2N]^-$ anions in ionic liquid phase. NO_3^- ions form H-

Table 3

Self-diffusion coefficients of $[C_4mim]^+$, of $[Tf_2N]^-$, R values, HOD, bond water, other species of ^{19}F detected in ILs phases.

Acid	$[HX]_{aq, init}$ M	D_+ ($m^2\ s^{-1}$)	D_- ($m^2\ s^{-1}$)	R	D_{HOD} ($m^2\ s^{-1}$)	$D_{bond\ water}$ ($m^2\ s^{-1}$)	$D_{other\ ^{19}F\ species}$ ($m^2\ s^{-1}$)
/	0	$3.75 \cdot 10^{-11}$	$2.63 \cdot 10^{-11}$	1.43	$4.97 \cdot 10^{-10}$	$3.72 \cdot 10^{-10}$	$^{a2.05 \cdot 10^{-9}}$ $^{a8.4 \cdot 10^{-10}}$ $^{3.11 \cdot 10^{-10}}$
DNO_3 in D_2O	1.99	$3.70 \cdot 10^{-11}$	$2.80 \cdot 10^{-11}$	1.32	Not measured		
	3.91	$4.20 \cdot 10^{-11}$	$3.20 \cdot 10^{-11}$	1.31			
HNO_3 in D_2O	4.00	$3.84 \cdot 10^{-11}$	$3.02 \cdot 10^{-11}$	1.27	$4.92 \cdot 10^{-10}$	$2.45 \cdot 10^{-10}$	$1.70 \cdot 10^{-9}$
	5.28	$4.54 \cdot 10^{-11}$	$3.64 \cdot 10^{-11}$	1.25	$4.85 \cdot 10^{-10}$	$2.21 \cdot 10^{-10}$	$^{a5.98 \cdot 10^{-10}}$
	7.22	$5.72 \cdot 10^{-11}$	$4.80 \cdot 10^{-11}$	1.19	$4.94 \cdot 10^{-10}$	$2.14 \cdot 10^{-10}$	$^{a1.89 \cdot 10^{-10}}$
					$^{b5.30 \cdot 10^{-11}}$		
HCl in D_2O	7.14	$3.32 \cdot 10^{-11}$	$2.60 \cdot 10^{-11}$	1.28	$4.82 \cdot 10^{-10}$	$1.71 \cdot 10^{-10}$	$^{a1.23 \cdot 10^{-9}}$

^a Indicates that the measurements of self-diffusion were difficult because of low intensity of signal.

^b See text.

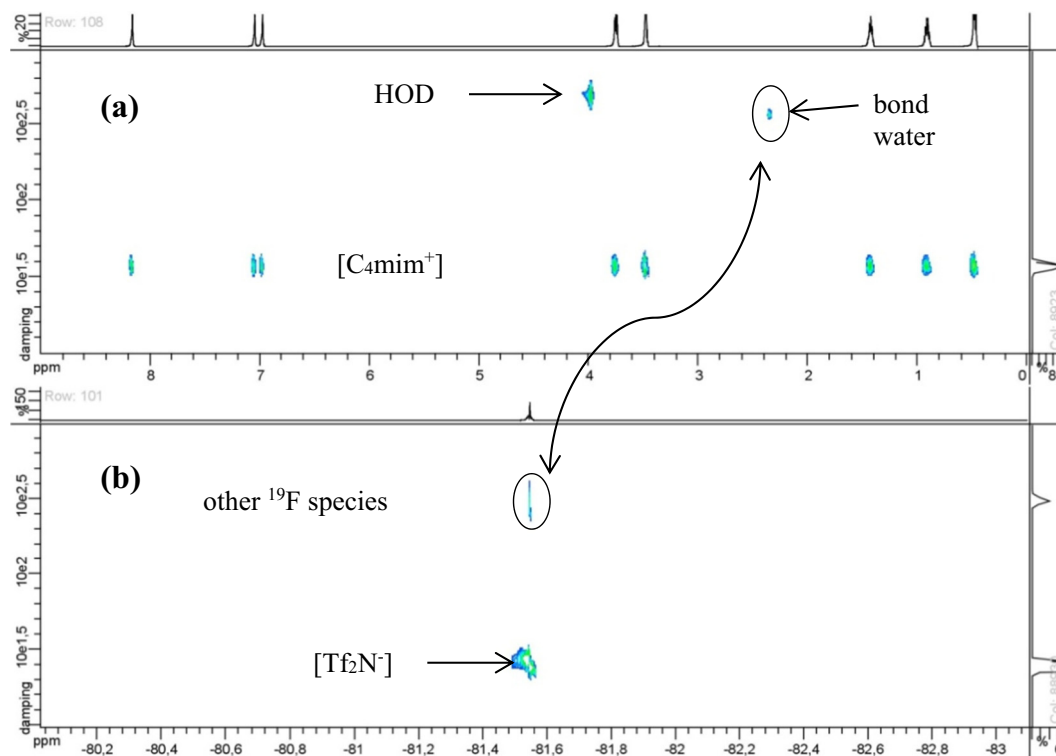


Fig. 7. ¹H DOSY spectra (a) and ¹⁹F DOSY spectra (b) of ionic liquid phase from the biphasic system HOD/[C₄mim][Tf₂N].

bonds with water molecules. In ionic liquid phase, nitrates amount is very low as compared to that of water which is extracted by ionic liquid as described in Table 1. The chemical species Tf₂N⁻⋯H—O—D⋯Tf₂N⁻ are no more observed while the HNO₃ is transferred in the ionic liquid phase. Free [Tf₂N⁻] are observed at [HNO₃]_{aq, init} = 4 M with a self-diffusion coefficient at 1.70·10⁻⁹ m² s⁻¹ and are not detected for [HNO₃]_{aq, init} > 4 M. Two self-diffusion coefficients are close to the background at 5.98·10⁻¹⁰ and 1.89·10⁻¹⁰ m² s⁻¹ and can be attributed at the species Tf₂N⁻⋯H—O—D⋯HNO₃. The exchange becomes faster as mole fraction of HOD increases in the ionic liquid phase [55,56]. Some species are detected by ¹⁹F DOSY NMR, but above [HNO₃]_{aq, init} = 4 M signals are closed to the background and it is not possible to conclude.

Self-diffusion assigned to bond water is a combination between self-diffusion of free water and the one of [Tf₂N⁻] anion. Assuming that the exchanges between bound and free species are fast and undetectable with the time scale of the NMR, the diffusion coefficients can be written by the following equation [26,57].

$$D_{\text{obs}} = f_{\text{free}} D_{\text{free}} + f_{\text{bound}} D_{\text{bound}} \quad (6)$$

In our case, D_{obs} is $D_{\text{bond water}}$, D_{free} is D_{HOD} and D_{bound} is D_{-} .

$$D_{\text{bondwater}} = f_{\text{free}} D_{\text{HOD}} + (1 - f_{\text{free}}) D_{-} \quad (7)$$

$$f_{\text{free}} = \frac{D_{\text{bond water}} - D_{-}}{D_{\text{HOD}} - D_{-}} \quad (8)$$

The free water fraction f_{free} is calculated from the self-diffusion coefficient measured in the ionic liquid phase according to Eq. (8). The values are 0.73, 0.47, 0.41, and 0.37 for each sample at [HNO₃]_{aq, init} = 0.00, 4.00, 5.28 and 7.22 M, and 0.32 with [HCl]_{aq, init} = 7.14 M. So, bond water fraction increases with the total water amount increasing in the ionic liquid phase. This, favors the formation of hydrogen bonds and exchange with the free water molecules as described by Eq. (5) which is rationalized by the deshielding of proton from water linked (see Fig. 2).

At the same HOD chemical shift at ca. 4 ppm one other self-diffusion coefficient measured at 5.30·10⁻¹¹ m² s⁻¹ (circle in red in Fig. SI 7) at [HNO₃]_{aq, init} = 7.22 M in deuterated phase. These signals reveal NOE effects which interfere with diffusion measurement [58,59]. That means free water molecules are closed to the [C₄mim⁺] cations with distance <5 Å.

The [C₄mim⁺] cation and [Tf₂N⁻] anion diffusions coefficients tend towards a similar value as shown by the decrease of ratio R. The water amount in ionic liquid phase increases with the partial transfer of HNO₃. Water induces a standardization of the [C₄mim⁺] cation and [Tf₂N⁻] anion diffusion coefficients. Water is an inclusion inside the frame build by [C₄mim⁺] cations and [Tf₂N⁻] anions [26,46,60] as shown in Fig. 5. The frame of [C₄mim][Tf₂N] ionic liquid begins to dismantle by the transfer of HOD, H₃O⁺ and nitrates. Anions [Tf₂N⁻] are solvated by water molecules and cations [C₄mim⁺] are placed in the second solvation sphere [22,61].

4. Conclusion

[C₄mim][Tf₂N] ionic liquid contacted with acidic aqueous solutions, extracts water and to a lesser extent H₃O⁺ and nitrates ions. A part of water is quasi free and moves ca. ten faster than [C₄mim⁺] cations and [Tf₂N⁻] anions. Free water can be considered as an inclusion in the frame of [C₄mim][Tf₂N] ionic liquid. The other part of water is bonded with [Tf₂N⁻] anions by its oxygen from sulfonyl group. As the amount of water, H₃O⁺ and nitrates ions increases in the [C₄mim][Tf₂N] ionic liquid phase, the bond water fraction increases. The network is little by little dismantled. [C₄mim⁺] cations and [Tf₂N⁻] anions get closer and move together as the values of R shown. They gather to form aggregates which include bond water with [C₄mim⁺] cations and [Tf₂N⁻] anions, H₃O⁺, nitrates ions. These clusters are solvated by free water.

Acknowledgments

The authors thank Maurice Coppe for 1D NMR measurements and Dr Lionel Allouche (Université de Strasbourg, Institut de Chimie) for

fruitful discussions. Special thanks are due Dr Bruno Vincent for 2D NMR measurements and values advices.

Funding

The present work was financially supported by the Interdisciplinary Mission at the Centre National de la Recherche Scientifique of France (CNRS) in the frame of the project “PHENOM”.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2018.08.086>.

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